

# Evolution of Edge Computing: 2019–2026

## Transition (After Maël speaks):

*“Thank you, Maël. To understand why Edge is so powerful today, we need to look at the innovations made since 2019.”*

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### Point 1: 5G

#### Standalone (SA)

The first big change is **5G Standalone**. In 2019, 5G was just starting. Today, it provides the “backbone” for Edge. It offers ultra-low latency—less than **10 milliseconds**. Without this speed, real-time Edge operations simply wouldn’t work.

### Point 2: Edge

#### AI & TinyML

Second, we have **Edge AI**, sometimes called **TinyML**. In the past, we sent data to the Cloud to be analyzed. Now, we run **Small Language Models** (SLMs) directly on the local microchips. This means the device can “think” and make decisions instantly, even without an internet connection.

### Point 3:

#### Management & Orchestration

Finally, we have solved the problem of scale. Managing thousands of small, scattered servers is a nightmare. But thanks to innovations like **KubeEdge**, which is a specialized version of **Kubernetes**, we can now manage decentralized nodes as easily as a single data center.

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## Transition (To Camille):

*“Now, Camille will explain how these technologies contribute to environmental preservation.”*